Chapter 3: Phase estimation and factorization

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1. The phase estimation problem

The spectral theorem is an important fact in linear algebra. It works for unitary matrices.

Theorem:
Suppose U is a unitary matrix of dimension N.There exists an orthonormal basis $|\psi_1\rangle, \ldots, |\psi_N\rangle$ of vectors with complex numbers such that:

$$\lambda_1 = e^{2\pi i\theta_1}, \ldots, \lambda_N = e^{2\pi i\theta_N}$$

such that:

$$U = \sum_{k=1}^{N} \lambda_k |\psi_k\rangle\langle\psi_k|$$

Each vector $|\psi_k\rangle$ is an eigenvector of U that has an eigenvalue λ_k:

$$U|\psi_k\rangle = \lambda_k|\psi_k\rangle = e^{2\pi i\theta_k}|\psi_k\rangle$$

Phase estimation problem:
In this problem, we have two things:

1. A description of a unitary circuit of n qubits.
2. A quantum state $|\psi\rangle$ of n qubits.

Phase estimation problem:

> Input: A unitary circuit for an n-qubit operation with an n-qubit quantum state.

> Promise: $|\psi\rangle$ is an eigenvector of U. Output: An approximation of the number $\theta \in [0, 1)$ satisfying:

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$$

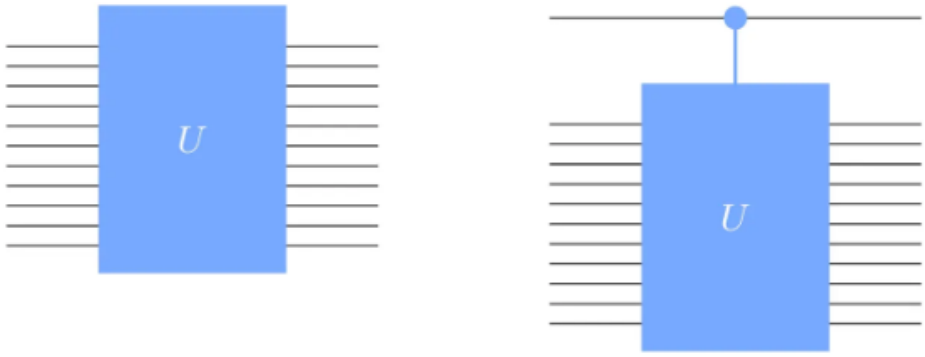
We can approximate θ by a fraction:

$$\theta \approx \frac{y}{2^m}$$

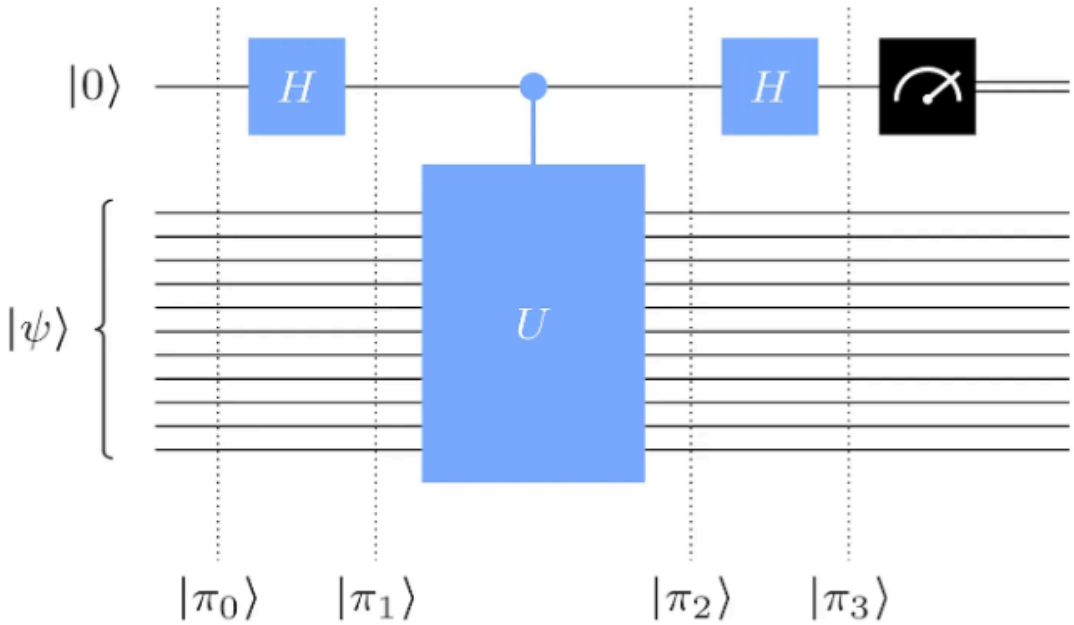
for $y \in \{0, 1, \ldots, 2^{m-1}\}$. (with modulo 1)

Warm up: using phase kickback

For a U circuit, you can create a controlled-U operator:



Consider the following circuit:



$$|\pi_0\rangle = |\psi\rangle|0\rangle$$

$$|\pi_1\rangle = \frac{1}{\sqrt{2}}|\psi\rangle|0\rangle + \frac{1}{\sqrt{2}}|\psi\rangle|1\rangle$$

$$|\pi_2\rangle = \frac{1}{\sqrt{2}}|\psi\rangle|0\rangle + \frac{1}{\sqrt{2}}(U|\psi\rangle)|1\rangle$$

$$|\pi_2\rangle = |\psi\rangle \otimes \ (\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}e^{2\pi i\theta}|1\rangle)$$

$$|\pi_3\rangle = |\psi\rangle \otimes \ (\frac{1+e^{2\pi i\theta}}{2}|0\rangle + \frac{1-e^{2\pi i\theta}}{2}|1\rangle)$$

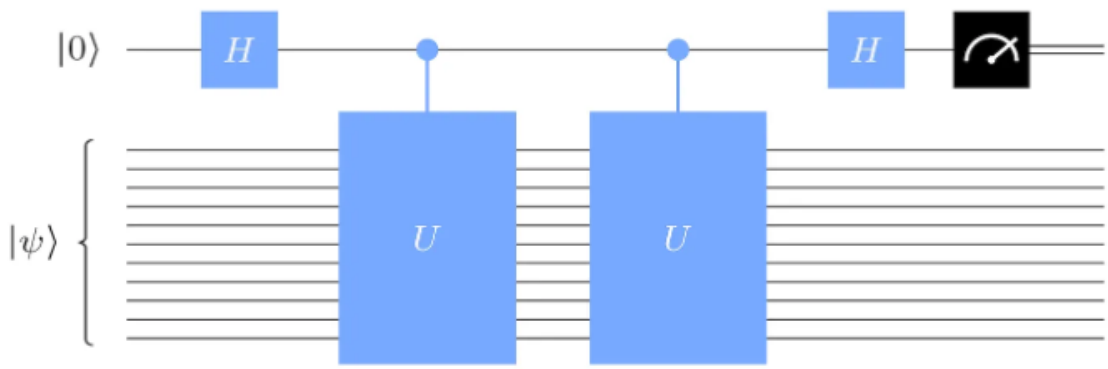
By measuring the probabilities of 0 and 1, we obtain:

$$p_0 = \left|\frac{1+e^{2\pi i\theta}}{2}\right| = cos^2(\pi\theta)$$

$$p_1 = \left|\frac{1-e^{2\pi i\theta}}{2}\right| = sin^2(\pi\theta)$$

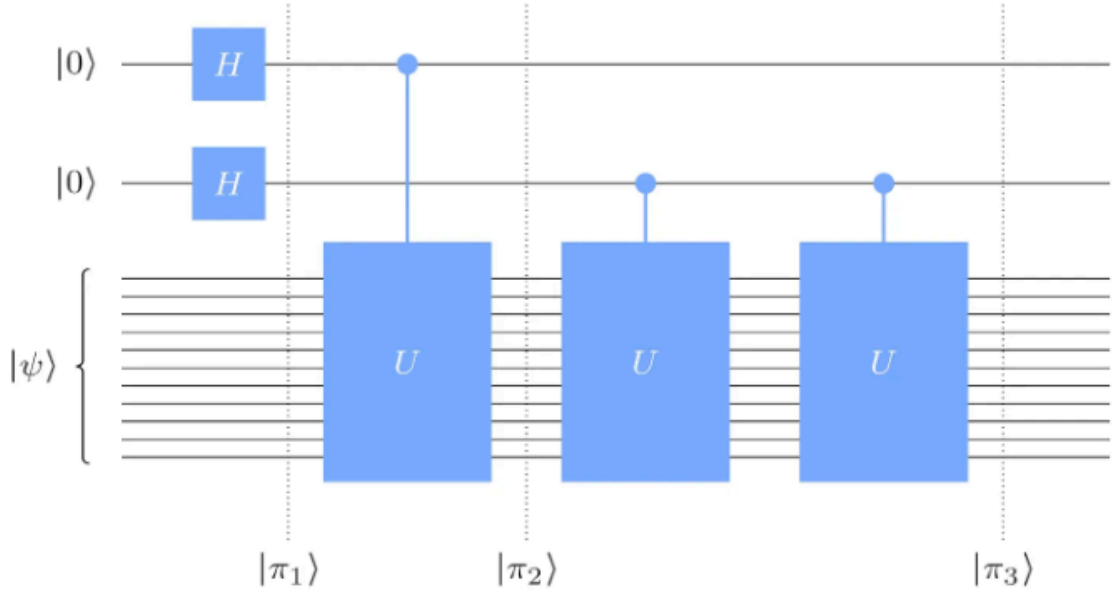
This is a way to obtain information about the eigenvalues of U. We obtain information about the phase.
Iterating the unitary operation

How can we find out more about θ?
One possibility is to apply the operation U twice to the circuit.



By applying the unitary matrix twice, we obtain the circuit in U^2. In this case, the applied phase is 2θ, and we obtain the desired phase by dividing the phase by 2.

Two control qubits:
Let us consider a circuit with one unitary operation controlled by one qubit and two other unitary operations controlled by another qubit.



$$|\pi_1\rangle = |\psi\rangle \otimes \sum_{a_0=0}^{1}\sum_{a_1=0}^{1}|a_1a_0\rangle$$

$$|\pi_2\rangle = |\psi\rangle \otimes \frac{1}{2}\sum_{a_0=0}^{1}\sum_{a_1=0}^{1}e^{2\pi i a_0\theta}|a_1a_0\rangle$$

$$|\pi_3\rangle = |\psi\rangle \otimes \frac{1}{2}\sum_{a_0=0}^{1}\sum_{a_1=0}^{1}e^{2\pi i(2a_1+a_0)\theta}|a_1a_0\rangle$$

Let $x = 2a_1 + a_0$. The factor 2 of a_1 comes from applying the circuit twice to a single qubit. We then have:

$$|\pi_3\rangle = |\psi\rangle \otimes \frac{1}{2}\sum_{x=0}^{3}e^{2\pi i x\theta}|a_1a_0\rangle$$

What can we know about θ? We can assume that we have a promise that $\theta = \frac{y}{4}$ for $y \in \{0, 1, 2, 3\}$. Can we know which one it is?

Let's define a 2-qubit state for each probability.

$$|\phi_y\rangle = \frac{1}{2}\sum_{x=0}^{3}e^{2\pi i\frac{xy}{4}}|a_1a_0\rangle$$

$$|\phi_0\rangle = \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle + \frac{1}{2}|2\rangle + \frac{1}{2}|3\rangle$$

$$|\phi_1\rangle = \frac{1}{2}|0\rangle + \frac{i}{2}|1\rangle - \frac{1}{2}|2\rangle - \frac{i}{2}|3\rangle$$

$$|\phi_2\rangle = \frac{1}{2}|0\rangle - \frac{1}{2}|1\rangle + \frac{1}{2}|2\rangle - \frac{1}{2}|3\rangle$$

$$|\phi_3\rangle = \frac{1}{2}|0\rangle - \frac{i}{2}|1\rangle - \frac{1}{2}|2\rangle + \frac{i}{2}|3\rangle$$

These four vectors are orthonormal; they perfectly describe a projection of measurements.

$$\{|\phi_0\rangle\langle\phi_0|, |\phi_1\rangle\langle\phi_1|, |\phi_2\rangle\langle\phi_2|, |\phi_3\rangle\langle\phi_3|\}$$

The unitary matrix whose columns are $|\phi_0\rangle, |\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle$ has this action:

$$V|y\rangle = |\phi_y\rangle, \quad \forall y \in \{0, 1, 2, 3\}$$

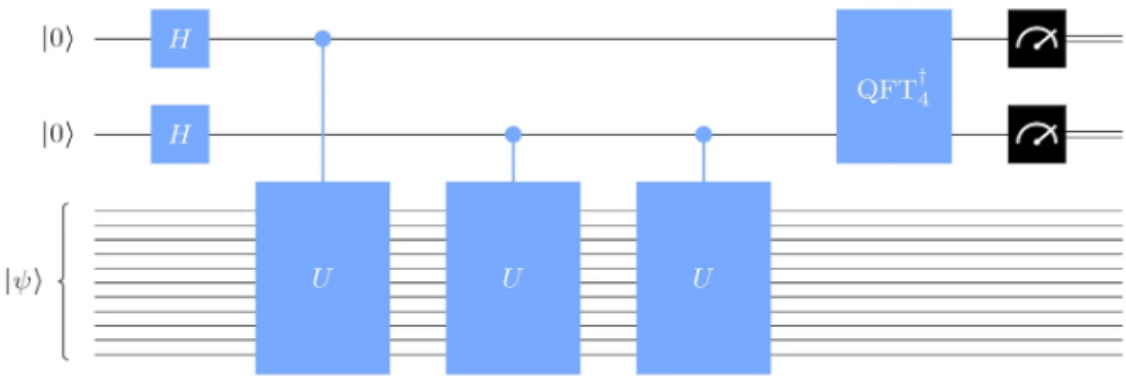
inverse of V in the standard basis:

$$V^\dagger|\phi_y\rangle = |y\rangle, \quad \forall y \in \{0, 1, 2, 3\}$$

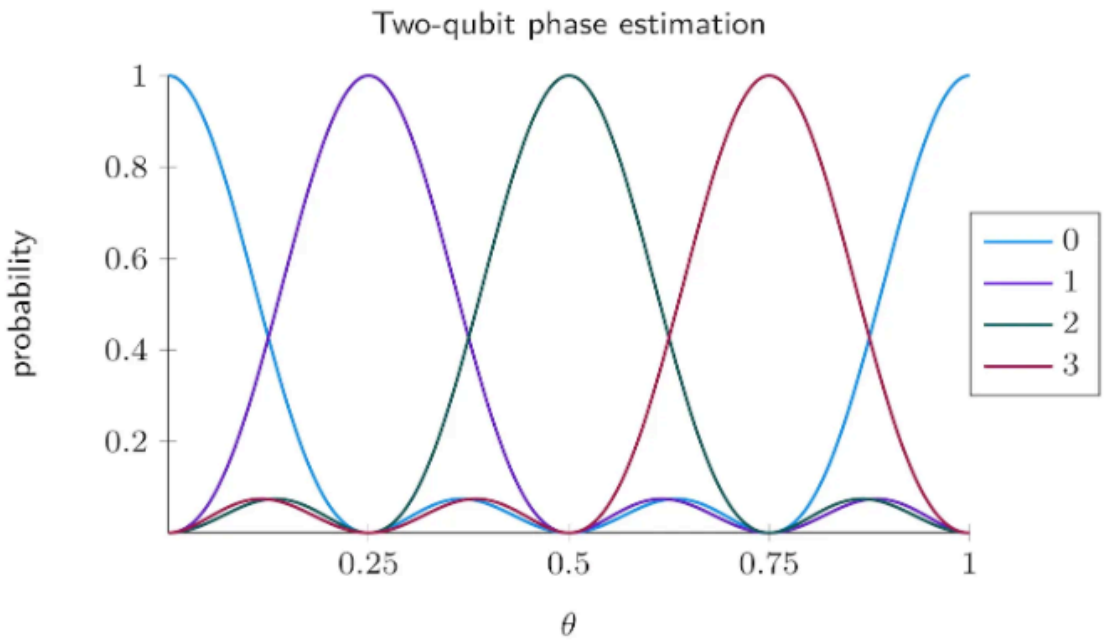
The resulting matrix corresponds to a discrete Fourier transform matrix (dimension 4):

$$V = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

This type of matrix is called a quantum Fourier transform. The complete circuit for learning about $y \in \{0, 1, 2, 3\}$.



The probability of output when the circuit is turned on:



According to the graph, the probability of obtaining the most accurate measurement of x is above 40%. Indeed, for values of $\theta = 3/8$, we arrive at the trough or the superposition of two curves, which corresponds to the minimum probability of measuring the information.

Quantum Fourier Transform: The quantum Fourier transform is defined for every positive integer N:

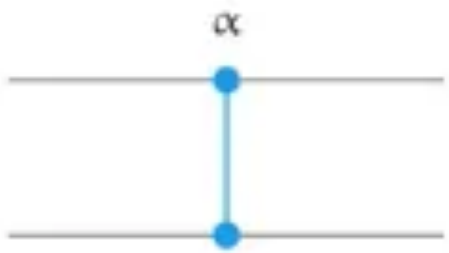
$$QFT_N = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} e^{\frac{2\pi i x y}{N}} |x\rangle\langle y|$$

$$QFT_N|y\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} e^{\frac{2\pi i x y}{N}} |x\rangle$$

Practical notation:

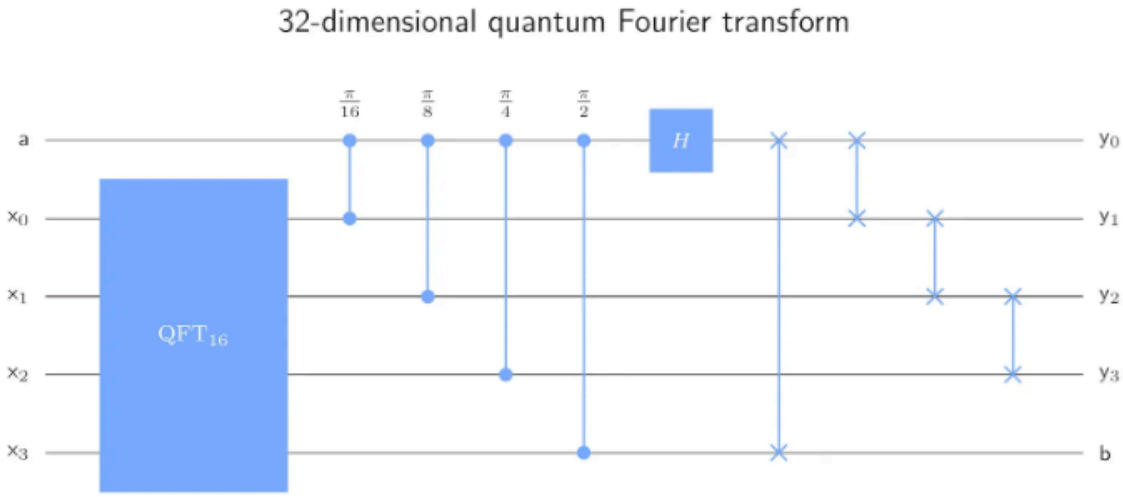
$$w_N = e^{\frac{2\pi i}{N}} = cos(\frac{2\pi}{N}) + i * sin(\frac{2\pi}{N})$$

The implementation of the quantum Fourier transform uses controlled-phase gates:



$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\alpha} \end{pmatrix}$$

The implementation is recursive in nature. For example, here we have a 32-bit QTF circuit:



Cost analysis: Let s_m be the number of gates needed for m qubits.

- For $m = 1$, a Hadamard gate is required
- For $m \geq 2$, the following gates are required:
- s_{m-1} gates for QFT on $m - 1$ qubits
- $m - 1$ controlled phase gates
- $m - 1$ swap gates
- 1 Hadamard gate

$$s_m = \begin{cases} 1 & \text{if } m = 1 \\ s_{m-1} + 2m - 1 & \text{if } m \geq 2 \end{cases} \tag{1}$$

By recurrence, we have a relationship of the form:

$$s_m = \sum_{k=1}^{m} (2k - 1) = m^2$$

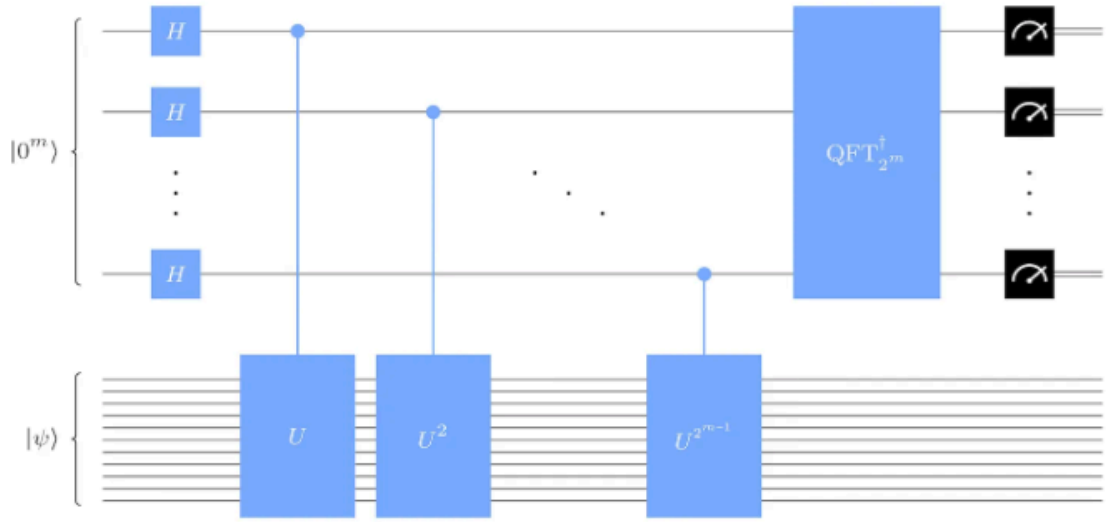
Note:

The number of door swaps can be reduced.The approximation of QFT_{2^m} can be done at a lower depth.

General procedure for phase estimation:

Warning:

If each U^k operation is used by repeating a controlled operation U k times, increasing the number of control qubits m comes at a high cost.



expression of the system upstream of the measurement:

$$|\pi\rangle = |\psi\rangle \otimes \frac{1}{2^m} \sum_{u=0}^{2^m-1} \sum_{x=0}^{2^m-1} e^{2\pi i x(\theta - y/2^m)} |y\rangle$$

Probaility to measure y:

$$p_y = \left| \frac{1}{2^m} \sum_{x=0}^{2^m-1} e^{2\pi i x(\theta - y/2^m)} \right|^2$$

The more qubits there are, the greater the precision.

<u>Best approximation</u>:

> Let's assume that $y/2^m$ is the best approximation of θ:
>
> $$\left| \theta - \frac{y}{2^m} \right| \leq 2^{-(m-1)}$$
>
> Then the probability of measuring y is highly relative:
>
> $$p_y \geq \frac{4}{\pi^2} \approx 0.405$$

<u>Worst approximation</u>:

> Suppose there is a best approximation of θ between $y/2^m$ and θ :
>
> $$\left| \theta - \frac{y}{2^m} \right| \geq 2^{-m}$$
>
> Then the probability of measuring y is very low:
>
> $$p_y \leq \frac{1}{4}$$
>
> To obtain an approximation of $y/2^m$, it is similar to satisfying

> $$\left|\theta - \frac{y}{2^m}\right| < 2^{-m}$$

We can run the estimation procedure using m qubits several times and take y to be the mode of the output. The eigenvector is not modified by the procedure and can be reused as many times as necessary.

3. Integer factorization by phase estimation

The order Finding problem:
For each positive integer N, we define:

$$\mathbb{Z} = \{0, 1, \ldots N-1\}$$

We define the modulo N on $\mathbb{Z}_N$

We say that the elements $a \in \mathbb{Z}_N$ are special when $gcd(a, N) = 1$, i.e.:

For any element $a \in \mathbb{Z}_N^* = \{a \in \mathbb{Z}_N : gcd(a, N) = 1\}$

Fact:

For any $a \in \mathbb{Z}_N^*$, there exists a positive integer k such that $a^k = 1$. The smallest k is called the order of a in $\mathbb{Z}_N^*$.

order Finding problem:

> Input: Positive integers a and N with gcd(a,N) = 1
> Output: The smallest positive integer r such that $a^r = 1(\mathrm{mod\ N})$

No classical algorithm can solve this problem.

To connect the problem with phase estimation, we will consider a set $\mathbb{Z}_N$.

For an element $a \in \mathbb{Z}_N^*$, we define the following operation:

$$M_a|x\rangle = |ax\rangle \ \ (\forall x \in \mathbb{Z}_N)$$

This is a unitary operation, because $gcd(a, N) = 1$.

The main idea:
The eigenvalues of M_a are strongly connected to the order of a. By approximating certain eigenvalues with sufficient precision using phase estimation, we are able to calculate the order.
Here is an eigenvector of M_a:

$$|\psi_0\rangle = \frac{|1\rangle + |a\rangle + \ldots + |a^{r-1}\rangle}{\sqrt{r}}$$

r is the order of a. The associated eigenvalue is 1:

$$M_a|\psi_0\rangle = \frac{|a\rangle + |a^2\rangle + \ldots + |a^r\rangle}{\sqrt{r}} = \frac{|a\rangle + |a^2\rangle + \ldots + |a^{r-1}\rangle + |1\rangle}{\sqrt{r}} = |\psi_0\rangle$$

To identify more eigenvectors, we can use the notation:

$$\omega_r = e^{2\pi i/r}$$

Here is another eigenvector of M_a:

$$|\psi_1\rangle = \frac{|1\rangle + \omega_r^{-1}|a\rangle + \ldots + \omega_r^{-(r-1)}|a^{r-1}\rangle}{\sqrt{r}}$$

by calculation:

$$M_a|\psi_1\rangle = \omega_r|\psi_1\rangle$$

in general,

$$M_a|\psi_j\rangle = \omega_r^j|\psi_j\rangle$$

Suppose that $|\psi_1\rangle$ is in a quantum state. We can learn about r by performing the following operations:

1. Use phase estimation on the state $|\psi_1\rangle$ and a quantum circuit implemented with M_a.
2. The output $2^m/y$ is rounded to the nearest integer

$$\text{round}(\frac{2^m}{y}) =< \left\lfloor \frac{2^m}{y} + \frac{1}{2} \right\rfloor$$

How much precision is needed to correctly determine r?

$$\left| \frac{2^m}{y} - \frac{1}{r} \right| \leq \frac{1}{2N^2} \Rightarrow \text{round}(\frac{2^m}{y}) = r$$

By choosing $m = 2lg(N) + 1$ in phase estimation, a good estimate can be obtained.

Suppose we have a vector $|\psi_j\rangle$ in a quantum state for a random choice of $j \in \{0, \ldots, r-1\}$. We can hope to learn about j/r in the following way:

1. Perform phase estimation on the state $|\psi_j\rangle$ and a quantum circuit implementing M_a. The output of the circuit gives an approximation of $y/2^m \approx j/r$
2. Among the fractions u/v whose lowest term satisfies $u, v \in \{0, \ldots, r-1\}$ and $v \neq 0$, the output that is closest to $y/2^m$. This can be done efficiently using the continued fraction algorithm.

How much precision is needed to correctly determine $u/v = j/r$?

$$\left|\frac{2^m}{y} - \frac{1}{r}\right| \leq \frac{1}{2N^2} \Rightarrow \frac{u}{v} = \frac{j}{r}$$

Choosing $m = lg(N) + 1$ for the phase estimation allows such an approximation.

We may be unlucky: j has a common factor with r.

In this case, if we draw samples independently for $j \in \{0, 1, \ldots r-1\}$ chosen uniformly, we can recover r with a high probability by calculating the least common multiple of the observed values of v.

<u>Implementation</u>:

Cost of the function $n = lg(N)$

- Cost for each controlled unitary

> M_a has a cost of $O(n^2)$
> We need to implement M_a^k for each $k = 1, 2, 4, 8, \ldots, 2^{m-1}$. Each M_a^k can be implemented as follows:
>
> > - Calculate $b = a^k \pmod N$
> > - Use a circuit for M_b The cost of implementing $M_b = M_a^k$ is $O(n^2)$

- m Hadamard gates: $O(n)$
- m unitary control operations: $O(n^3)$
- Quantum Fourier Transform: $O(n^2)$

Total cost: $O(n^3)$

Reminder: we must obtain one of the eigenvectors $\psi_0, \psi_1, \ldots, \psi_{r-1}$.

Solution: replace the eigenvector ψ with $|1\rangle$

This works because:

$$|1\rangle = \frac{|\psi_0\rangle + |\psi_1\rangle + \ldots + |\psi_{r-1}\rangle}{\sqrt{r}}$$

The output is the same if we choose $j \in \{0, \ldots, r-1\}$ uniformly and use $|\psi\rangle = |\psi_j\rangle$

Factorization through order finding:

The following method allows us to find a factor of N with a probability of at least $1/2$, to predict that N is odd and not prime.

Factor finding method:

1. Choose $a \in \{2, \ldots, N-1\}$ randomly.

2. Calculate $d = gcd(a, N)$. If $d \geq 2$ then the output is d and stop.
3. Calculate the order r of a modulo N
4. If r is even, then calculate $d = gcd(a^{1/2} - 1, N)$. If $d \geq 2$, then d is the output and stop
5. If this step is reached, the method has failed.

Main Idea:

> 1. By the definition of order, we know that $a^r = 1(mod\ N)$
>
> $$a^r = 1(mod\ N) \Leftrightarrow \text{N divides } a^r - 1$$

> 2. If r is even, then:
>
> $$a^r - 1 = (a^{1/2} + 1)(a^{1/2} - 1)$$

Every prime number that divides N must therefore divide $(a^{1/2} + 1)$ or $(a^{1/2} - 1)$. For a random number a, at least one prime number of N divides $(a^{1/2} - 1)$.

4. Implementation on Qiskit

In [3]:
```
from qiskit.quantum_info import Statevector
from qiskit.quantum_info.operators import Operator
from qiskit.visualization import plot_histogram
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator

from math import pi, cos, sin
```

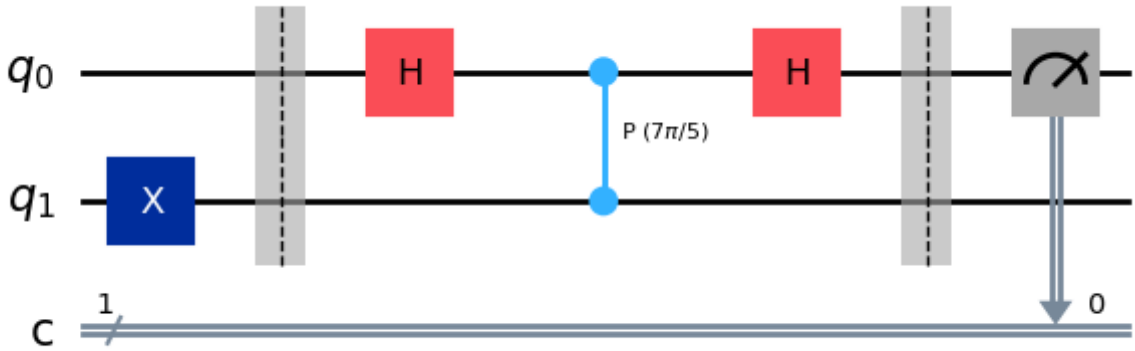
In [2]:
```
psi1 = Statevector([cos(pi / 8), sin(pi / 8)])
psi2 = Statevector([cos(5 * pi / 8), sin(5 * pi / 8)])

# When given a Statevector input, the Operator function returns the outer
# product of that state vector with itself — or, in other words, the
# product of the vector times its conjugate transpose.

H = Operator(psi1) - Operator(psi2)
display(H.draw("latex"))
```

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

Using the phase kickback

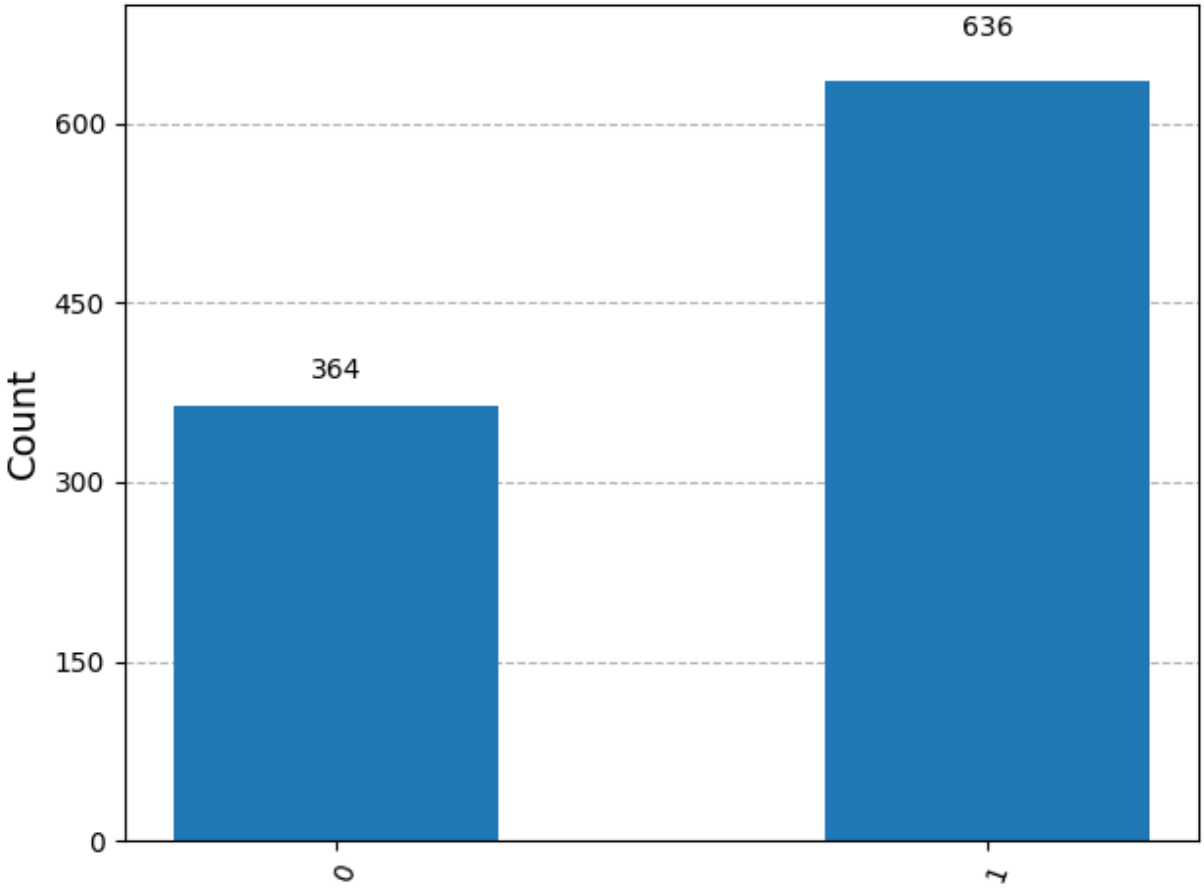
In [4]:
```
theta = 0.7  # Can be changed to any value between 0 and 1
qc = QuantumCircuit(2, 1)

# Prepare the eigenvector, which is the |1> state
```

```
qc.x(1)
qc.barrier()

# Implement the estimation procedure
qc.h(0)
qc.cp(2 * pi * theta, 0, 1)
qc.h(0)
qc.barrier()

# Perform the final measurement
qc.measure(0, 0)

# Draw the circuit
display(qc.draw(output="mpl"))
```



In [5]:
```
result = AerSimulator().run(qc, shots = 1000).result()
stat = result.get_counts()
plot_histogram(stat)
```

Out[5]:



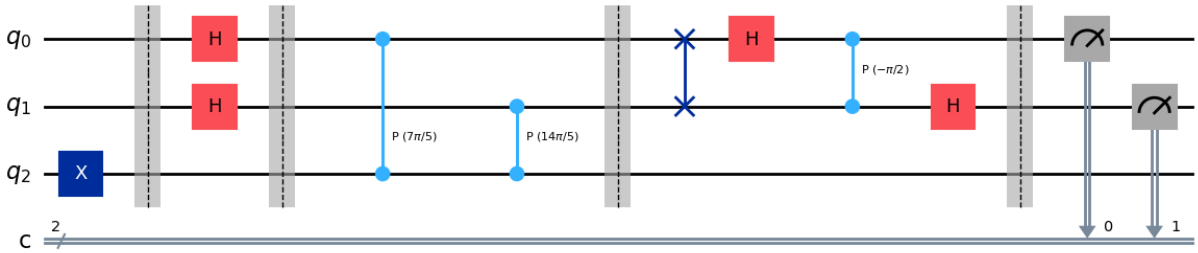
In [6]:
```
print(f"cos(pi * {theta})^2 * nb_shot = {cos(pi * theta) ** 2*1000:.4f}")
print(f"sin(pi * {theta})^2 * nb_shot = {sin(pi * theta) ** 2*1000:.4f}")
```

```
cos(pi * 0.7)^2 * nb_shot = 345.4915
sin(pi * 0.7)^2 * nb_shot = 654.5085
```

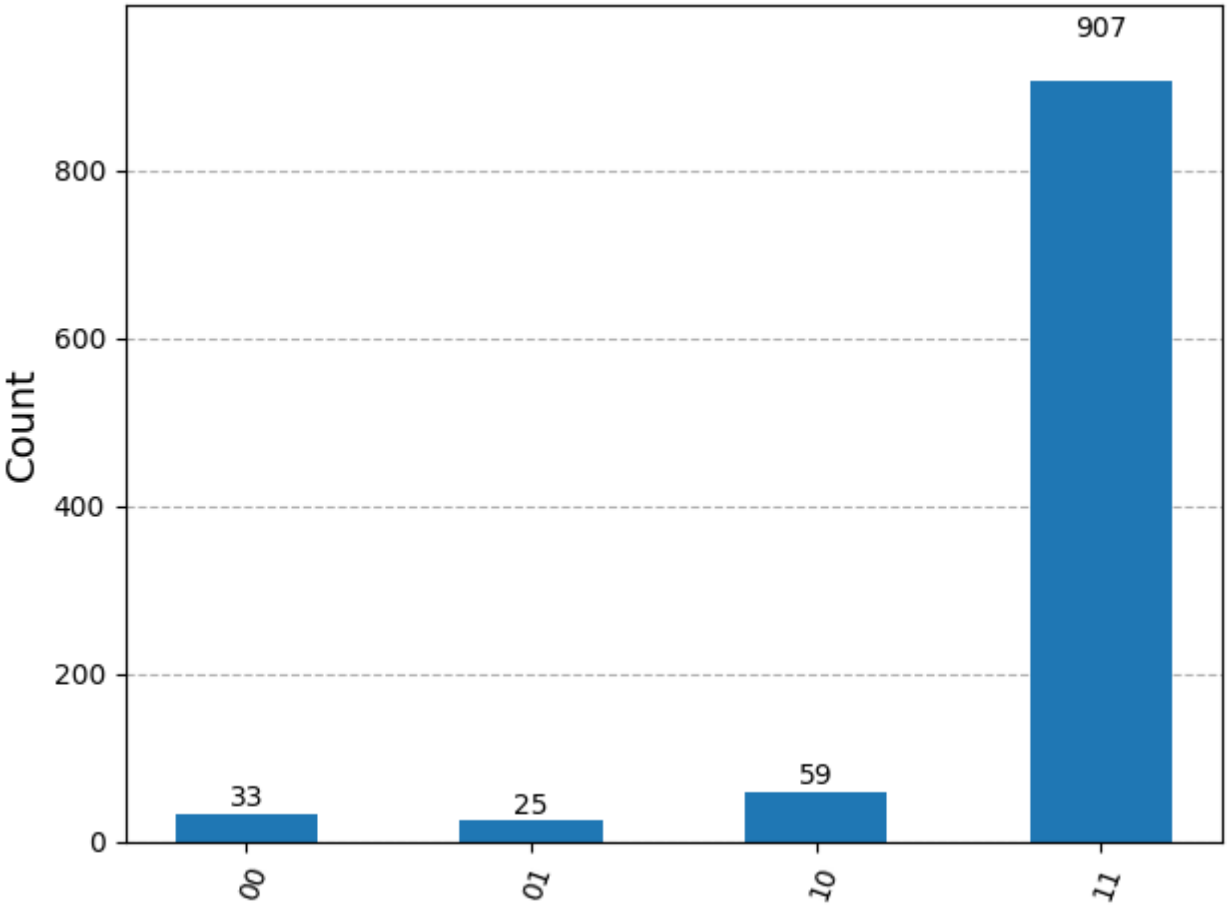
Quantum Fourier Transform (dim 4)

In [7]:
```
thetat = 0.7
qc = QuantumCircuit(3,2)

# prepare the eigenvector
qc.x(2)
qc.barrier()

# the initla hadamard gates
qc.h(0)
qc.h(1)
qc.barrier()

# the controlled unitary gates
qc.cp(2*pi*theta, 0, 2)
qc.cp(2*pi*(2*theta), 1, 2)
qc.barrier()

# implementation of the inverse of the two-qubit QFT
qc.swap(0,1)
qc.h(0)
qc.cp(-pi/2,0,1)
qc.h(1)
```

```
qc.barrier()

qc.measure([0,1],[0,1])

display(qc.draw("mpl"))
```



In [8]:
```
result = AerSimulator().run(qc).result()
stat = result.get_counts()
display(plot_histogram(stat))
```



In [10]:
```
qc = QuantumCircuit(3)

# prepare the eigenvector
qc.x(2)
qc.barrier()

# the initla hadamard gates
qc.h(0)
qc.h(1)
qc.barrier()
```

```
# the controlled unitary gates
qc.cp(2*pi*theta, 0, 2)
qc.cp(2*pi*(2*theta), 1, 2)
qc.barrier()

# implementation of the inverse of the two-qubit QFT
qc.swap(0,1)
qc.h(0)
qc.cp(-pi/2,0,1)
qc.h(1)
qc.barrier()
sv = Statevector.from_instruction(qc)
sv.draw("latex")
```

Out[10]:

$$(0.1727457514 + 0.0561284971i)|100\rangle + (0.0675438064 + 0.1325621841i)|101\rangle + (-0.07725424$$

Quantum Fourier Transform

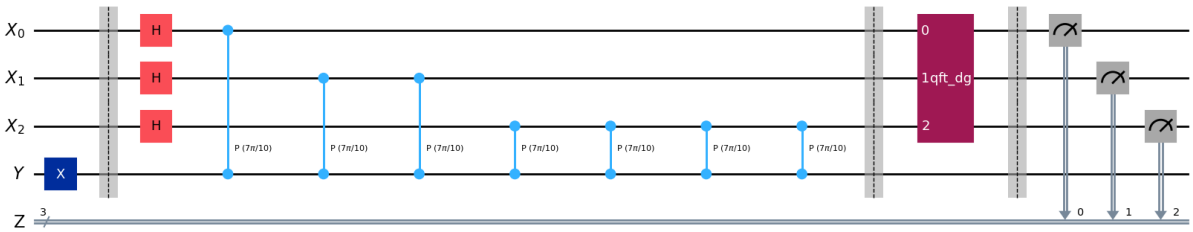
In [13]:

```
from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister
from qiskit_aer import AerSimulator
from qiskit.circuit.library import QFTGate
```

In [15]:

```
theta = 0.35
m = 3# nombr de qubits de control

control = QuantumRegister(m, name = "X")
target = QuantumRegister(1, name = "Y")
output = ClassicalRegister(m, name="Z")
qc = QuantumCircuit(control, target, output)

# prepare the eigenvector
qc.x(target)
qc.barrier()

# perform phase estimation
for index, qubit in enumerate(control):
    qc.h(index)
    for _ in range(2**index):
        qc.cp(2*pi*theta, qubit, target)
qc.barrier()

# perform the inverse quantum Fourier Transform
qc.compose(
    QFTGate(m).inverse(),
    inplace = True
)
qc.barrier()

qc.measure(range(m), range(m))
qc.draw("mpl")
```

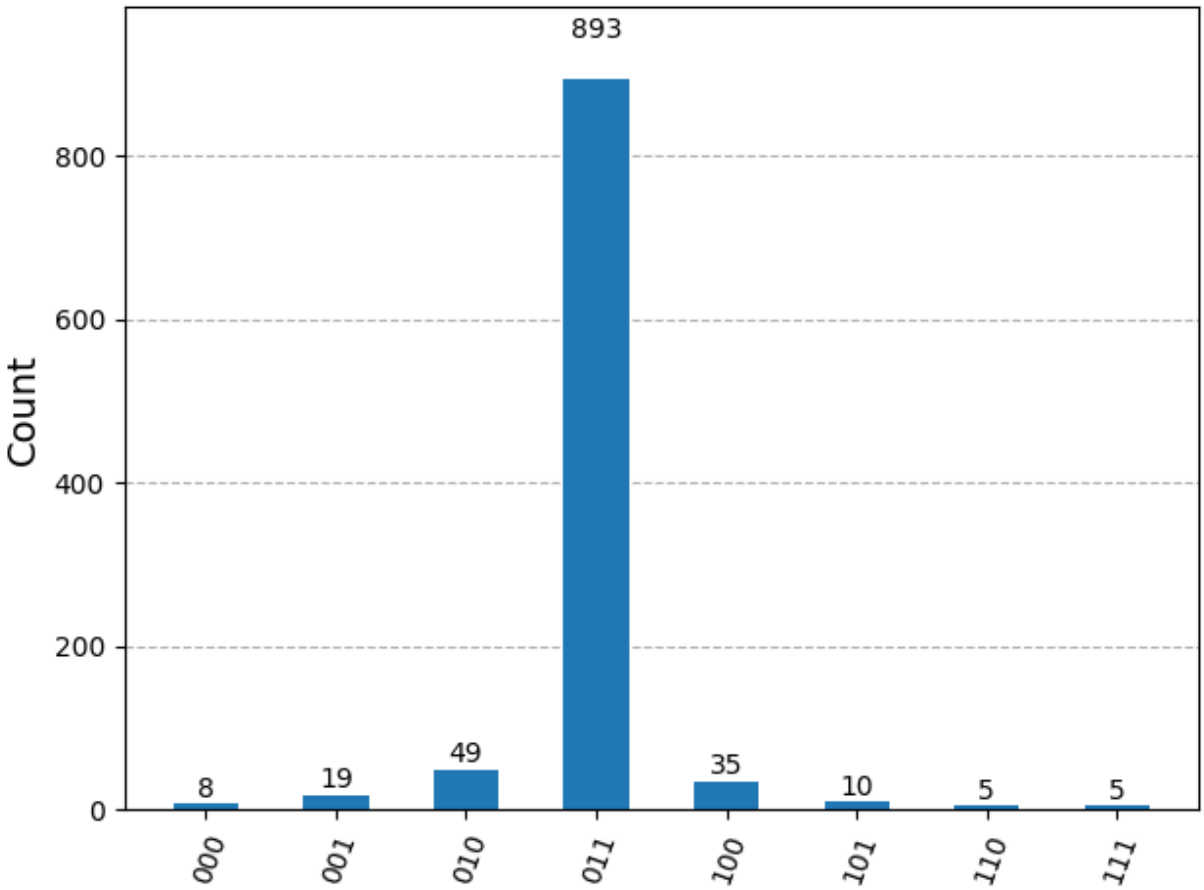
Out[15]:



In [17]:

```
result = AerSimulator().run(qc.decompose()).result()
stat = result.get_counts()
plot_histogram(stat)
```

Out[17]:



In [18]:

```
most_probable = int(max(stat, key=stat.get),2)

print(f"Most probable output: {most_probable}")
print(f"Estimated theta: {most_probable/2**m}")
```

```
Most probable output: 3
Estimated theta: 0.375
```

Shor Algorithm

In [19]:

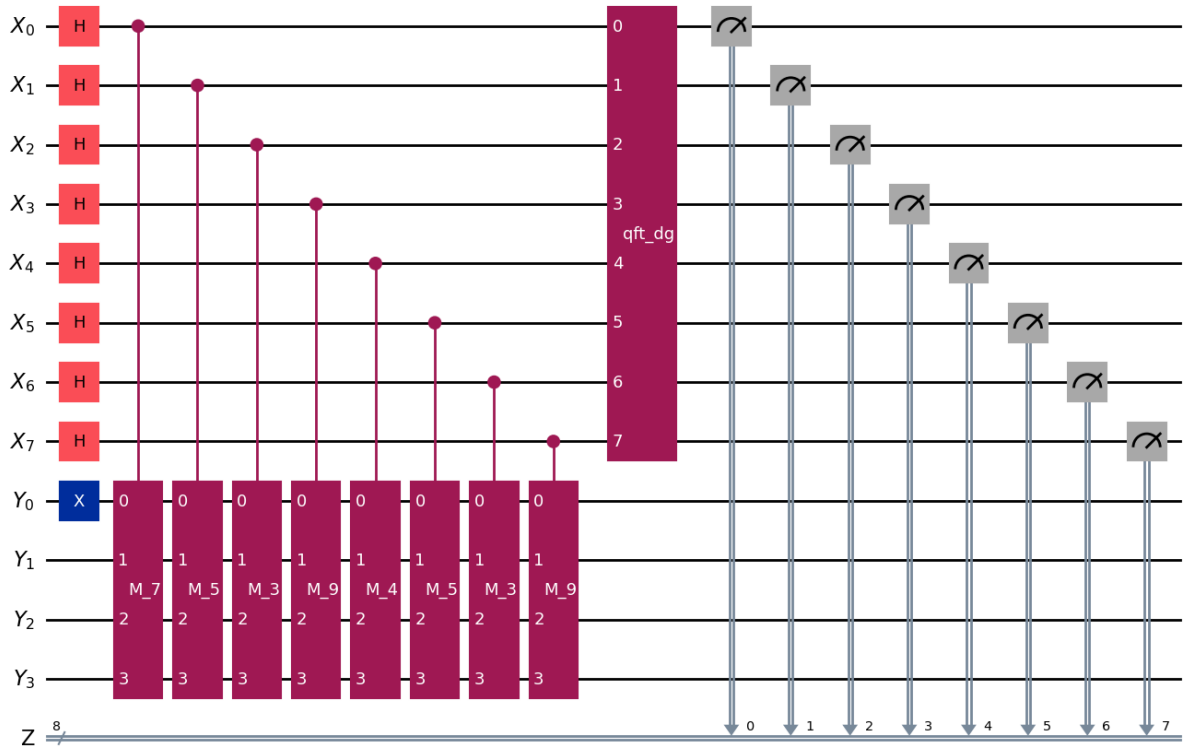
```
from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister, tr
from qiskit.quantum_info.operators import Operator
from qiskit.circuit.library import UnitaryGate
from qiskit.circuit.library import QFTGate
from qiskit_aer import AerSimulator
import numpy as np
from math import gcd, floor, log
from fractions import Fraction
```

```python
import random
```

In [24]:
```python
def mod_mult_gate(b,N):
    # M_b function for an element of B in Zn
    if gcd(b,N)>1:
        print(f"Error: gcd({b},{N}) > 1")
    else:
        n = floor(log(N-1,2))+1
        U = np.full((2**n, 2**n),0)
        for x in range(N): U[b*x % N][x] = 1
        for x in range(N, 2**n): U[x][x] = 1
        G = UnitaryGate(U)
        G.name = f"M_{b}"
        return G

def order_finding_circuit(a,N):
    # Create the circuit for order finding
    if gcd(a,N)>1:
        print(f"Error: gcd({a},{N}) > 1")
    else:
        n = floor(log(N-1,2))+1
        m = 2*n

        control  =QuantumRegister(m, name = "X")
        target = QuantumRegister(n, name="Y")
        output = ClassicalRegister(m, name="Z")
        circuit = QuantumCircuit(control, target, output)

        # initialize the target register to the state \1>
        circuit.x(m)

        # Add hadamard gates and controlled versions of the multiplicatio
        for k, qubit in enumerate(control):
            circuit.h(k)
            b = pow(a,2**k,N)
            circuit.compose(
                mod_mult_gate(b,N).control(),
            qubits = [qubit] + list(target), inplace = True)

        # apply the inverse QFT to the control register
        circuit.compose(
            QFTGate(m).inverse(),
            qubits = control, inplace = True)

        #measure the control register
        circuit.measure(control, output)
        return circuit
```

In [25]:
```python
# Example
order_finding_circuit(7,11).draw("mpl")
```

Out[25]:



In [26]:

```
# to run the circuit on simulator
def find_order(a,N):
    if gcd(a,N)>1:
        print(f"Error: gcd({a},{N}) > 1")
    else:
        n = floor(log(N-1,2))+1
        m = 2*n
        circuit = order_finding_circuit(a,N)
        transpiled_circuit = transpile(circuit, AerSimulator())

        while True:
            result = AerSimulator().run(transpiled_circuit, shots = 1, me
            y = int(result.get_memory()[0],2)
            r = Fraction(y/2**m).limit_denominator(N).denominator
            if pow(a,r,N)==1 : break
        return r
```

In [27]:

```
N = 11
a = 7
print(f"The order of {a} modulo {N} is {find_order(a,N)}.")
```

```
The order of 7 modulo 11 is 10.
```

In [ ]:

```
# To run the circuit on real device
def find_order(a,N):
    if gcd(a,N)>1:
        print(f"Error: gcd({a},{N}) > 1")
    else:
        n = floor(log(N-1,2)) + 1
        m = 2*n
        circuit = order_finding_circuit(a,N)
        transpiled_circuit = transpile(circuit,AerSimulator())

        service = QiskitRuntimeService()
```

```
        backend = service.least_busy(min_num_qubits=127)
        pass_manager = generate_preset_pass_manager(optimization_level=3,

        isa_circuit = pass_manager.run(transpiled_circuit)

            y = int(result.get_memory()[0],2)
            r = Fraction(y/2**m).limit_denominator(N).denominator
            if pow(a,r,N)==1: break
        return r
```